

Remote Patient Monitoring System

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Technical project brief

A desktop application for reviewing real-time physiological signals, patient records, and alerts through a structured, extensible clinical software architecture.

Problem

Clinical monitoring systems need a coherent view of live vital signs, historical information, and abnormal events while retaining traceability across patient records.

System architecture

Java Swing client $\longleftrightarrow$ REST API $\longleftrightarrow$ PostgreSQL database

Implemented functionality

- Multi-patient monitoring interface with live ECG and respiration waveform visualisation.
- Patient onboarding, persistent clinical records, and threshold-based visual and audio alerts.
- Event acknowledgement and stored alert causes to support clinical review and traceability.
- Digital-twin visualisation module as an extensible representation of physiological state.

System map

Presentation layer	Java Swing desktop interface for multi-patient views and live waveform display
Application layer	REST services coordinating clinician, patient, vital-sign, and alert workflows
Data layer	PostgreSQL persistence for structured patient data, historical records, and alert events
Quality controls	JUnit test suite and Gradle build workflow

Project outcome

The completed prototype delivered a coherent clinical workflow from patient onboarding to live monitoring, alert acknowledgement, and review of stored health information. It demonstrated how a modular architecture can support a future monitoring platform without coupling the interface to one hardware device.

Technical tools

Java Swing REST APIs PostgreSQL Gradle JUnit Digital health software